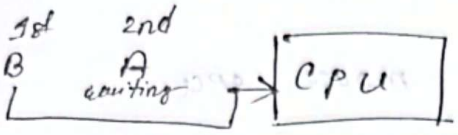
২২ slide important

03-02-2026

How cpu executes Orders:



=> যাকে আগে পাবে Run করবে, পরের টা ৩৪ ন

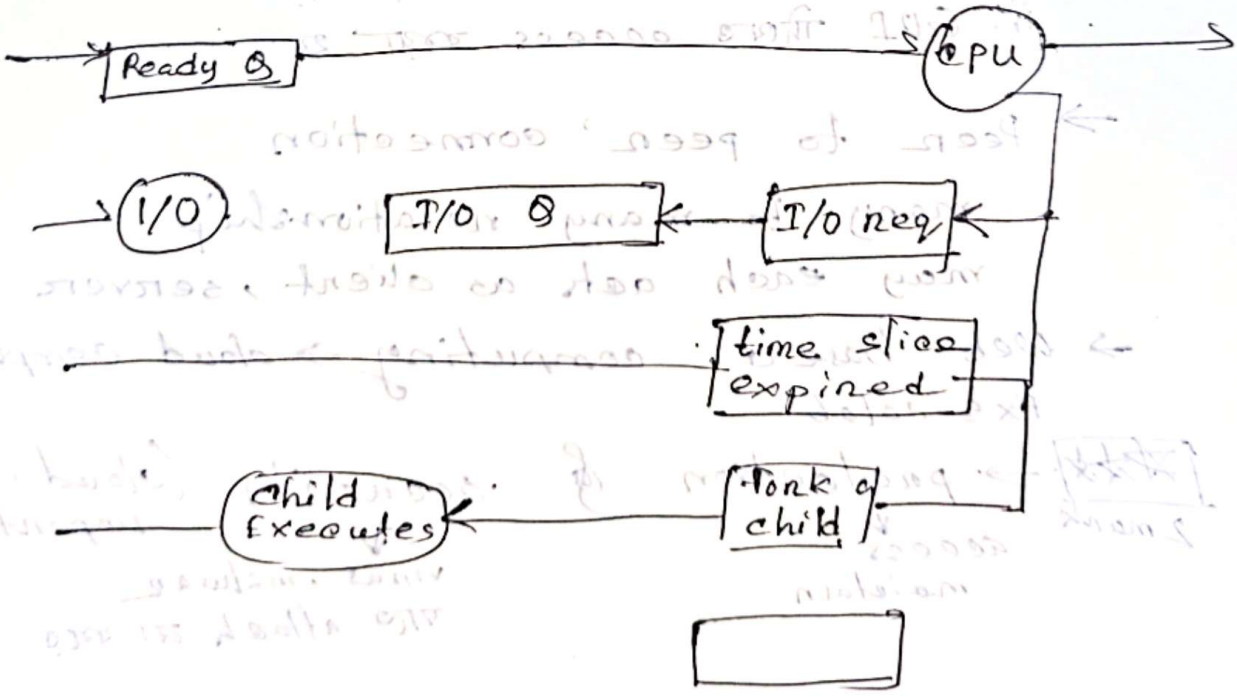
=>

কোন process execute করবে সেটা বুঝাতে জন্য → CPU scheduling Algorithm

.o file → program ready to execute

Process scheduling:

→ job q
→ Read q
→ Device q



Types of schedulers :

→ Long term schedulars (job schedulin
(sort / যেটা আগে Read Queue তে পাঠাবে).

→ Short-term schedulers (CPU sched
(which process should be executed next and allocates CPU)

বাদ Addition of Medium term scheduling

Depending on time Scheduler Process — 2 types

i. I/O bound process : ২টা সমস্যা যোগ করবে। (input নিতে time নেয়)

ii. CPU bound process : ~~memory~~ ~~com~~ input নেয়ার পর compute করতে time নেয় (process করতে time নেয়)

Ex : 3D image process, resize, overlap coloring, pixel change

CPU burst কি

=> একটা process continuous stage এ একটানা Run করলে সেটাই CPU burst

Ex : 1st burst, 2nd burst
(বাস 35)

scheduling Algorithm → 10 marks

Goals :

i. maximize utilization = কম সময়ে সর্বোচ্চ use করতে পারি

ii. minimize turn around time :

(turn around time : total computation time, আসার time থেকে শেষ হওয়ার time)

iii. minimize avg response time :
কোন process এ long time এর করতে না হয়

iv. Fairness : all process must be treated equally. negardless of arrival T.

v. Minimize overhead : process টা তাড়াতাড়ি run করবে যাতে upper bound, lower bound maintain করে,

Arrival time : সব process same time এ আসে না, time stamp অনুযায়ী আসে

Burst time : Execution time, যা

Total Burst : একটা process multiple time run হয়ে execute হয়েছে,

তাহলে 1st burst time → 3

2nd " " → 4

∴ Total → 7

Waiting time : Execute করার জন্য যতক্ষণ wait করতে হয়।

wait time = Arrival time − executes time

Turn Around Time :

CPU শুরু থেকে শেষ কি সময়

= completion time − arrival time

= waiting time + burst time

completion time : → total burst time

time required to complete its executions.

Types of Algorithms :

→ pre-emptive : নতুন প্রসেস priority পেলেই

→ Non-pre emptive : priority based sorting নেই ,

Operating System

Multiprocessors

- Multiprocessing : 2 or more cpu
  - → Assymmetrical Multiprocessing
    Ex: সব গুলো cpu always কাজ করে না,
    1 টা cpu → lead cpu, hierarchy থাকে
  - → Symmetrical Multiprocessing
    EX: all equal

- cluster System : → sharing storage via network
  (combination of machines) → provides high availability

  cluster computing :

- Computing Environments
  - → client server computing
    i: terminal দিয়ে access করা হয়
    ii: GUI দিয়েও access করা হয়
  - → Peer to peer connection
    - many to many relationship
    may each acts as client, server or both.
  - → web based computing → cloud computing
    Ex: colab

Symm clustering → সমান কাজ
asymm clustering

2 mark → protection & security (cloud computing important term)
protection → access maintain
security → virus, malware থাকে attack না করে

System Call

fork : child process create করা
(Recursion নয় অতো)

Services provide by system calls

i. process creation and management
ii

Types of sys call

(I) process control

(II) file mana

(III)

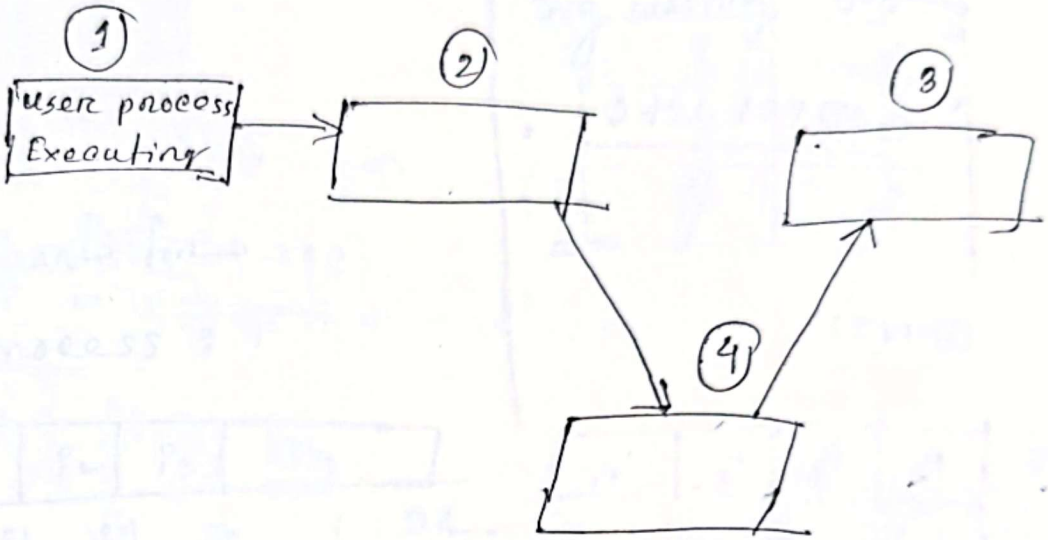
(IV)

(V)

System call ⇒ আমরা যখন kernel এর request করব

User mode = high level execution এর জন্য

kernel mode = calculation



Next week CT-01 (lec 1 → 7)